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J Lat Am Geriat Med. 2025;11(2):35-42

THE JOURNAL OF LATIN AMERICAN GERIATRIC MEDICINE

ORIGINAL ARTICLE

Psychosocial factors associated with frailty in older people living in Mexico City with the use of telemedicine: a pilot study

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Abstract

Background: *The global elderly population is growing rapidly, and the number of people aged 65 or older is expected to double by 2050. In Mexico, this trend is also evident, posing a significant challenge for social and health policies. Frailty in this population group is an increasing problem that requires multidisciplinary attention.* **Objective:** *To explore the association between sociodemographic, clinical, and psychosocial factors and frailty in older adults, using telemedicine as a tool for remote assessment.* **Material and methods:** *A study was conducted in which sociodemographic, clinical, and psychosocial factors were evaluated in older adults using telemedicine. Variables such as age, hypercholesterolemia, history of depression, and cognitive impairment, among others, were analyzed.* **Results:** *Significant associations were found between frailty and age ($p < 0.003$), hypercholesterolemia ($p < 0.021$), history of depression ($p < 0.005$), and cognitive impairment ($p < 0.024$). These results highlight the relevance of these factors in the development of frailty.* **Conclusion:** *The findings emphasize the importance of multidisciplinary approaches to prevent and manage frailty in older adults, particularly in vulnerable populations. In addition, it is suggested that early detection and interventions through telemedicine could improve quality of life and reduce disability in this growing demographic group.*

Keywords: *Geriatrics. Elderly. Frailty. Telemedicine. Aging.*

Factores psicosociales asociados a la fragilidad en adultos mayores residentes en la Ciudad de México con el uso de la telemedicina: un estudio piloto

Resumen

Antecedentes: *La población mundial de adultos mayores está experimentando un crecimiento acelerado, y se proyecta que el número de personas de 65 años o más se duplique para 2050. En México, este fenómeno también es evidente, lo que representa un desafío significativo para las políticas sociales y de salud. La fragilidad en este grupo poblacional es un problema creciente que requiere atención multidisciplinaria.* **Objetivo:** *Explorar la asociación entre factores sociodemográficos, clínicos y psicosociales con la fragilidad en adultos mayores, utilizando la telemedicina como herramienta para la evaluación remota.* **Material y métodos:** *Se realizó un estudio en el que se evaluaron factores sociodemográficos, clínicos y psicosociales en adultos mayores mediante el uso de telemedicina. Se analizaron variables como la edad, la presencia de hipercolesterolemia, antecedentes de depresión y deterioro cognitivo, entre otros.* **Resultados:** *Se identificaron asociaciones significativas entre la fragilidad y la edad ($p < 0{,}003$), la hipercolesterolemia ($p < 0{,}021$), los antecedentes de depresión ($p < 0{,}005$) y el deterioro cognitivo ($p < 0{,}024$). Estos resultados destacan la relevancia de estos factores en el desarrollo de la fragilidad.* **Conclusiones:** *Los hallazgos subrayan la importancia de enfoques multidisciplinarios para prevenir y manejar la fragilidad en adultos mayores, especialmente en poblaciones vulnerables. Además, se sugiere que la detección temprana y las intervenciones a través de la telemedicina podrían mejorar la calidad de vida y reducir la discapacidad en este grupo demográfico en crecimiento.*

Palabras clave: *Geriatría. Adultos mayores. Fragilidad. Telemedicina. Envejecimiento.*

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Date of reception: 29-01-2025
Date of acceptance: 02-05-2025
DOI: 10.24875/LAGM.25000002

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INTRODUCTION

The older adult population is increasing rapidly worldwide, and data from the "World Social Report 2023" presented by the United Nations[1] show that the world's population aged 65 and over is expected to double from 761 million in 2021 to 1.6 billion in 2050, with even faster growth in the 80+ age group. Thus, in 2021, one out of every ten people was 65 years of age or older, and by 2050, it is expected to be one out of every six. In Mexico, according to data from the Consejo Nacional de Población[2], the population aged 65 years or older in the second decade of this century was 14.4 million, representing 11.2% of the total. It is estimated that by 2030, this demographic group will reach 20.6 million, constituting 15% of the total population, and by 2070, it is expected to reach 48.3 million, representing 34.2% of the total population.

Although the increase in life expectancy has been seen as an improvement in health systems worldwide, the number of years lived in good health or at least without severe disability has increased, indicating that middle-aged men live longer than women, who experience a higher prevalence of morbidity and disability[1]. In this sense, disability is often preceded by frailty, which is a state characterized by a decline in the ability to respond to demands, caused by a decrease in functional reserve[3].

One of the most accepted conceptualizations at the moment is that frailty is a biological syndrome resulting from a decrease in the internal homeostasis of the individual, which will cause a reduction in resistance to stressful events that increase the vulnerability of the individual, such as hospitalization[4]. This definition includes five criteria: unintentional weight loss, low energy, muscle weakness, slow gait, and low physical activity. Older adults with three or more criteria are defined as "frail," those with one or two criteria are "pre-frail," and those who do not meet any criteria will be either "non-frail" (or "robust" to some).

A second definition is based on the multifactorial accumulation of deficits[3]. In this model, frailty is the consequence of a sum of deficits acquired over time, including several aspects, such as cognitive, functional, and social, which goes beyond only the physical aspects included in the previous definition. In this broader definition, other relevant components are also included, such as sensory aspects, mood or cognitive alterations, and even socioeconomic variables, all of which can lead to a state of greater vulnerability in the elderly.

Frailty is strongly associated with mortality, this being the most recognized adverse outcome[5,6]. Moreover, it has shown good predictive ability for various health-related outcomes in different settings, such as hospitals, communities, and nursing homes[7]. In Mexico, frailty prevalence varies significantly, as evidenced by a systematic review and meta-analysis[8] of 16 studies, reported rates ranging from 8.6% to 50.9%, with a pooled prevalence of 31.2% (95% confidence interval [CI]: 24.9-37.4%) and high heterogeneity (98.7%). Regional studies such as the one conducted in Western Mexico[9] found a frailty prevalence of 33% in older adults, with key components including fatigue (42%), slow walking speed (35%), and unintentional weight loss (34%).

In this framework, it is essential to identify psychosocial factors associated with frailty in the older adult population, taking advantage of telemedicine as an innovative tool for assessment and continuous monitoring. The World Health Organization defines telemedicine as "the provision of health services by health professionals, where distance is a critical factor, using information and communication technologies to exchange valid information for the purposes of diagnosis, treatment and prevention of disease and injury, research and evaluation, and to facilitate the continuing education of health professionals, with the aim of safeguarding the health of individuals and communities"[10]. In Mexico, telemedicine has been adopted across various specialties–including neuropsychology[11], neurology[12], oncology[13], diabetes management[14], and tobacco cessation[15]–with diverse levels of success and accessibility.

During the COVID-19 pandemic, its use expanded rapidly, highlighting both its potential and structural limitations, for instance, while studies in teleneuropsychology reveal high adoption rates among clinicians (82%), they also point to key challenges such as limited access to technology, lack of standardized tools, and lower perceived diagnostic certainty compared to in-person care[11]. Similarly, the use of telemedicine platforms such as WhatsApp and phone consultations–driven by their accessibility–has underscored the trade-offs between technological reach and clinical depth[12]. Despite these barriers, telemedicine has shown promise in reaching vulnerable populations. A study with 178 Mexican older adults found that 92.7% use social media daily, primarily for social connection (49.4%) and health information (44.9% reported making health decisions based on online content). These findings suggest that digital

engagement–when properly guided–could complement telemedicine by fostering social support and health literacy, both critical for mitigating frailty's psychosocial drivers (e.g., isolation, depression)[16].

MATERIAL AND METHODS

Study design and population

This study was conducted remotely through teleconsultation. The inclusion criteria focused on individuals aged 60 years or older, residing in Mexico City, who provided informed consent over the telephone before their initial assessment. The study population consisted of community-dwelling, non-institutionalized older adults.

Participants who met the inclusion criteria and agreed to participate were assessed between January and July 2024. The teleconsultations lasted approximately 50-60 min and included an *ad hoc* questionnaire with medical and sociodemographic information such as name, age, gender, occupation, schooling, medications taken and history of illness, cognitive assessment using the Mini–Mental State Examination over the Telephone (t-MMSE[17]), the Montreal Cognitive Assessment (MoCA BLIND/Telephone[18]), and self-reported assessment of level of functioning using indices for basic and instrumental activities of daily living[19,20]. Frailty was assessed using the FRAIL questionnaire[21] to categorize patients (robust, pre-frail, and frail), nutritional status (MNA-SF[22]) and, finally, symptoms of depression and anxiety were assessed using the geriatric depression scale (GDS-15)[23] and GAD-7[24] scales, respectively.

The exclusion criteria included individuals who were unable to understand the study, provide independent consent, or lacked an informant to corroborate collected information. Participants had the option to withdraw at any time and were encouraged to express any questions or concerns during the study period.

Recruitment

Recruitment for the study was carried out through the various social networks of the Instituto Nacional de Ciencias Médicas y Nutrición "Salvador Zubirán" (Fig. 1). The recruitment strategy was focused on identifying people who might be at risk of frailty or pre-frailty. To do so, an invitation was distributed that included a self-assessment questionnaire with key questions aimed at detecting signs of decreased strength and endurance, such as fatigue, difficulty climbing stairs, walking short distances, the presence of multiple chronic diseases, and significant weight loss. The results of the questionnaire determined the classification of participants into two categories: pre-frail or frail, which allowed their inclusion in the program.

The registration process was carried out through telephone lines and WhatsApp, providing accessibility and ongoing support to those interested.

Data analysis

The statistical analysis for this study was conducted using the Statistical Package for the Social Sciences (SPSS) software for Windows (SPSS Inc., Chicago, IL version 25.0). Descriptive statistics were employed to summarize the sociodemographic and clinical characteristics of the participants, including gender, age, educational attainment, marital status, and health-related factors such as comorbidities and history of depression or cognitive impairment.

The participants were categorized into two groups: non-fragile (n = 34) and pre-fragile/fragile (n = 37). Comparative analyses between these groups were performed using χ^2 tests to evaluate associations between frailty status and various factors, including age, hypertension, diabetes, hypercholesterolemia, depression, falls, and cognitive impairment. Statistical significance was set at $p < 0.05$.

RESULTS

Table 1 provides an overview of the sociodemographic characteristics of the participants in the study. Women are more represented in the sample, constituting 69.1%, while men make up 30.8%. In terms of educational attainment, the majority of participants (77.5%) have >10 years of formal education, whereas 22.5% have fewer than 10 years of education. In addition, 40.8% of the participants report being married, while 59.2% are not married.

The participants were divided into two groups: non-fragile (n = 34) and pre-fragile and fragile (n = 37). Significant differences were observed in some characteristics, such as older age, where the group aged 71-80 years presented a higher prevalence of frailty (74.2%) compared to the group aged 60-70 years (38.7%) and the group over 81 years (22.2%) ($p < 0.003$). Likewise, a significant association was identified between frailty and hypercholesterolemia

Table 1. Sociodemographic characteristics of the participants

Variables	(n = 71) (%)
Age range	
60-70 years old	31 (43.66)
71-80 years old	31 (43.66)
Over 81 years old	9 (12.67)
Gender	
Male	22 (30.8)
Female	49 (69.1)
Years of education	
< 10 years	16 (22.5)
> 10 years	55 (77.5)
Marital status	
Married	29 (40.8)
Not married	42 (59.2)

Sociodemographic characteristics of the participants (n = 71). Data are presented as frequencies and percentages, showing the distribution of age range, gender, years of education, and marital status among the study participants.

($p < 0.021$) and with a history of depression ($p < 0.005$) and cognitive impairment ($p < 0.024$).

It stands out that the history of depression and cognitive impairment was more frequent in the pre-frail and frail group, with 85.7% ($p < 0.005$) and 70.8% ($p < 0.024$), respectively, compared to the non-frail group.

The relationship between frailty and depression was also supported by the results of the geriatric depression scale-15, where 87.5% of participants classified as frail or pre-frail had scores indicative of depression ($p < 0.001$). No significant differences were found in the prevalence of frailty in relation to history of smoking, alcoholism, and/or comorbidities such as systemic arterial hypertension, diabetes, or history of falls in the last year. These results underscore the importance of addressing both physical and psychosocial factors in the assessment of frailty, particularly those related to mental health and cognitive impairment, which show a strong association with frailty status in this population. The data are shown in detail in table 2.

A logistic regression (frail/pre-frail and non-frail) revealed that hypercholesterolemia (odds ratio [OR] = 7.13, 95% CI [1.10, 46.49], p = 0.040) and depression (OR = 14.12, 95% CI [1.69, 118.32], p = 0.015) were significant predictors of frailty. However, the small sample size (n = 71) and wide CIs suggest that these findings need to be replicated in larger studies.

DISCUSSION

The accelerated aging of the population in Mexico predicts that by 2050, one in six Mexicans will be 65 years of age or older[2]. In this context, frailty emerges as a key syndrome that seriously affects the quality of life of older adults, increasing disability and dependency, requiring a comprehensive approach that considers both biological and psychosocial factors[3,4].

This investigation examined frailty syndrome in Mexico City's elderly population through telemedicine platforms. Analysis revealed important correlations between frailty status and several variables: increased age, elevated cholesterol levels, depressive symptoms, and cognitive decline. These observations underscore the importance of combined clinical strategies that address both biological markers and mental health components, especially in high-risk groups.

The study indicates that telemedicine shows particular promise for initial frailty screening and basic evaluation; however, researchers emphasize that comprehensive longitudinal data are needed to fully establish its reliability as a primary diagnostic method. The technology appears most advantageous for communities facing challenges in accessing conventional medical services, though the study's design (single-timepoint assessment with limited sample size) necessitates cautious interpretation of results.

These benefits of telemedicine could potentially enhance seniors' well-being while decreasing pressure on medical infrastructure[25,26]. Subsequent studies should focus on three key areas: durability of outcomes, implementation across diverse populations, and direct comparisons with traditional clinical assessments.

STRENGTHS

One of the main strengths of this study lies in the implementation of telemedicine, which allowed participants to be assessed remotely, eliminating geographical barriers and facilitating access to medical care[25]. This approach is particularly relevant in the Mexican context, where large distances and limited availability of resources in rural and remote areas can make access to in-person care difficult[26]. Telemedicine not only offers a practical solution to these challenges but also proves to be a valuable tool for the early detection and monitoring of conditions such as frailty syndrome in older adults.

Another crucial aspect is the ability of telemedicine to detect frailty early, allowing for timely interventions

Table 2. Clinical and psychosocial characteristics associated with frailty

Variables	Non-frail (n = 34) (%)	Pre-frail and frail (n = 37) (%)	p
Sex			
Male	14 (63.6)	8 (36.4)	0.075
Female	20 (40.8)	29 (59.2)	
Age			
60-70 years old	19 (61.3)	12 (38.7)	0.003
71-80 years old	8 (25.8)	23 (74.2)	
> 81 years old	7 (77.8)	2 (22.2)	
Alcoholism			
Yes	1 (25.0)	3 (75.9)	0.346
No	33 (49.3)	34 (50.7)	
Smoking			
Yes	1 (20.0)	4 (80.0)	0.195
No	33 (50.0)	33 (50.0)	
Polypharmacy			
Yes	10 (38.5)	16 (61.5)	0.227
No	24 (53.3)	21 (46.7)	
Diabetes			
Yes	4 (40.0)	6 (60.0)	0.590
No	30 (49.2)	31 (50.8)	
Hypercholesterolemia	4 (44.4)	13 (76.5)	0.021
Yes	30 (55.6)	24 (44.4)	
No			
Systemic arterial hypertension			
Yes	15 (45.5)	18 (54.5)	0.702
No	19 (50.0)	19 (50.0)	
CVD			
Yes	2 (100)	0 (0.0)	0.135
No	32 (46.4)	37 (53.6)	
Malnutrition			
Yes	3 (27.3)	8 (72.7)	0.137
No	31 (51.7)	29 (48.3)	
Obesity			0.885
Yes	5 (50)	5 (50)	
No	29 (47.5)	32 (52.5)	
History of depression			
Yes	2 (14.3)	12 (85.7)	0.005
No	32 (56.1)	25 (43.9)	
History of cognitive impairment			
Yes	7 (29.2)	17 (20.8)	0.024
No	27 (57.4)	20 (42.6)	
History of anxiety			
Yes	10 (34.5)	19 (65.5)	0.060
No	24 (57.1)	18 (42.9)	
Hypothyroidism			
Yes	2 (33.3)	4 (66.7)	0.456
No	32 (49.2)	32 (50.8)	
Falls in the past year			
Yes	9 (40.9)	13 (59.1)	0.430
No	25 (51.0)	24 (49.0)	
Densitometry			
Yes	17 (41.5)	24 (58.5)	0.205
No	17 (56.7)	13 (43.3)	
Fractures			
Yes	1 (16.7)	5 (83.3)	0.110
No	33 (50.8)	32 (49.2)	

(*Continues*)

Table 2. Clinical and psychosocial characteristics associated with frailty *(continued)*

Variables	Non-frail (n = 34) (%)	Pre-frail and frail (n = 37) (%)	p
Years of schooling			
< 10 years	4 (25.0)	12 (75.0)	0.037
> 10 years	30 (54.5)	25 (45.5)	
Osteoporosis			
Yes	5 (31.3)	11 (68.8)	0.130
No	29 (52.7)	26 (47.3)	
Osteopenia			
Yes	11 (36.7)	19 (63.3)	0.105
No	23 (56.1)	18 (43.9)	
GDS-15			
Yes	2 (12.5)	14 (87.5)	0.001
No	32 (58.2)	23 (41.8)	

Clinical and psychosocial characteristics associated with frailty. The table compares the distribution of variables between non-frail (n = 34) and pre-frail and frail (n = 37) participants, including sex, age, lifestyle factors, comorbidities, and psychological conditions. Significant associations ($p < 0.05$) were observed for age, hypercholesterolemia, history of depression, cognitive impairment, years of schooling, and GDS-15.
CVD: cardiovascular disease; GDS: geriatric depression scale.

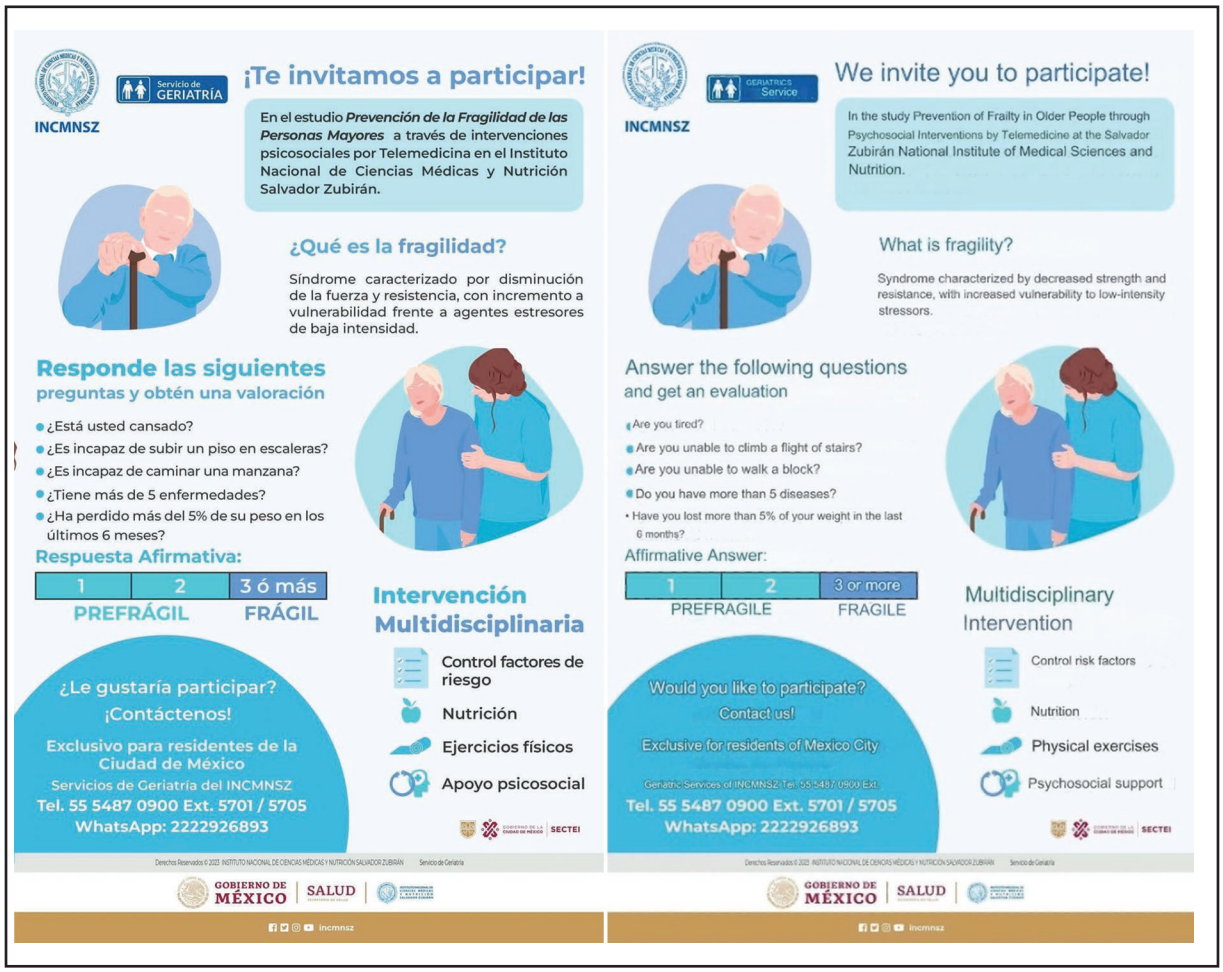

Figure 1. *Publication on social networks for participant recruitment.*

to be implemented before the condition progresses to greater disability[26]. This has significant implications for public health, as early intervention can considerably reduce the incidence of disability, improve the quality of life of older adults, and decrease the burden on health systems. The results of this study suggest that telemedicine could become an essential tool in the management of frailty, especially in vulnerable populations where access to care is limited.

Limitations

Despite the clear advantages offered by telemedicine, it is important to recognize that its dependence on technology may have excluded certain sectors of the population, particularly older people with limited digital literacy or without access to technological devices[27]. This potential bias could have influenced the study sample, favoring a population with better economic and technological resources, which may limit the generalizability of the findings. Evidence from telemedicine implementation in Mexico underscores these concerns: studies on teleneuropsychology and teleneurology revealed substantial barriers linked to technological limitations, including poor internet connectivity, reliance on smartphones with restricted functionality, and the lack of digital skills among both patients and providers[11,12]. In one study, nearly 30% of clinicians did not routinely obtain informed consent for remote evaluations, partly due to the absence of formal training and guidelines[11]. Furthermore, socioeconomic disparities–such as the digital divide between urban and rural areas (72% vs. 50% internet penetration, respectively)–further compound these challenges[11]. A common obstacle reported across various telemedicine programs in Mexico was patients' limited experience with communication technologies and the absence of suitable infrastructure, such as private spaces or reliable devices[13]. These limitations highlight the urgent need to develop inclusive strategies that not only ensure technological access but also actively promote digital literacy among older adults. Without such measures, telemedicine risks reinforce existing health inequities.

Another point to consider is the cross-sectional design of the study, which prevents establishing causal relationships between the identified factors and frailty. Although the observed associations are significant, longitudinal studies are required to evaluate how these factors may influence the development and progression of frailty over time. Longitudinal assessment would allow a better understanding of the underlying mechanisms and would offer a more solid basis for developing preventive interventions.

ACKNOWLEDGMENT

The authors would like to thank Dr. A. Ávila-Funes for his leadership in the development of this project and for his invaluable support for the use of telemedicine to address frailty syndrome in older adults in Mexico City. We also extend our thanks to Dr. I. Uscanga for her essential collaboration in the selection of the sample, which contributed significantly to the development of this research.

FUNDING

The authors declare that this project was funded by the Secretaría de Educación, Ciencia, Tecnología e Innovación of Mexico City, through the 2019 Call, under folio 1336c22.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

ETHICAL CONSIDERATIONS

Protection of humans and animals. The authors declare that this study was reviewed and approved by the Research Ethics Committee of the Instituto Nacional de Ciencias Médicas y Nutrición "Salvador Zubirán", under folio GER-698-12-19-1. All procedures performed in this protocol comply with applicable national and international ethical standards, in accordance with the Declaration of Helsinki. Respect for the principles of autonomy, beneficence, non-maleficence, and justice was guaranteed, ensuring the well-being of the participants in all phases of the study. All research subjects gave their informed consent before their inclusion in the study.

Confidentiality, informed consent, and ethical approval. The authors have followed their institution's confidentiality protocols, obtained informed consent from patients, and received approval from the Ethics Committee. The SAGER guidelines were followed according to the nature of the study.

Declaration on the use of artificial intelligence. The authors declare that no generative artificial intelligence was used in the writing of this manuscript.

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